THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 7 & 8: **How do Organisms Reproduce?, Heredity** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the testes, Student A says they produce testosterone; Student B says they produce sperms. Then: 1

(a) Both are correct **(b)** Only Student B is correct
(c) Only Student A is correct **(d)** Both are wrong

2. Which of these is a bisexual flower? 1

(a) hibiscus **(b)** papaya **(c)** watermelon **(d)** cucumber

3. Leishmania causes: 1

(a) AIDS **(b)** typhoid **(c)** kala-azar **(d)** malaria

4. The cutting and ligation of vas deferens is called: 1

(a) vasectomy **(b)** condom **(c)** copper-T **(d)** tubectomy

5. The structure that protects spores of Rhizopus is: 1

(a) flagellum **(b)** thick wall **(c)** pseudopodia **(d)** cilia

6. For pea plants, Student A says round seed is dominant over wrinkled; Student B says wrinkled is dominant. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

7. Which genotype will give short plants? 1

(a) Tt **(b)** tt **(c)** TT **(d)** all of these

8. In a dihybrid cross, the number of phenotypes in F2 is: 1

(a) 2 **(b)** 9 **(c)** 3 **(d)** 4

9. The number of autosomes in a human somatic cell is: 1

(a) 23 **(b)** 46 **(c)** 44 **(d)** 22

10. The gene for a hormone-producing enzyme being less efficient results in: 1

(a) no change **(b)** shorter plants **(c)** white flowers **(d)** taller plants

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is vegetative propagation practised for growing some types of plants? 2

12. A student says Amoeba reproduces by multiple fission. Justify or refute. 2

13. Name the scientist known as the father of genetics and the plant he used. 2

14. A student says acquired traits like muscle building are inherited. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain the events that follow fertilisation in a flower. 3

16. Why is the child's sex considered to be determined by the father? 3

17. What are STDs? Name two and suggest two preventive measures. 3

OR

Why is the reproductive system not fully developed at birth? What changes occur at puberty in general? 3

18. A cross between two tall plants gives 25% short plants. What are the genotypes of the parents? Show the cross. 3

OR

Write three features of Mendel's experimental method that made it successful. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe the human male reproductive system with its functions.
(ii) What is the role of hormones in it? 5

OR

(i) Explain fission, fragmentation and regeneration with examples.
(ii) Why can't complex organisms reproduce by regeneration? 5

20. (i) Explain sex determination in humans with a diagram.
(ii) Why is it wrong to blame women for the birth of daughters? 5

OR

(i) What is heredity?
(ii) What are genes and chromosomes?
(iii) How are traits expressed?
(iv) Give one example of a dominant trait in humans. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND